

● EASY

● ALGEBRA

Algebra

8 questions · ~12 min

26

Q1

ALGEBRA

$$s = 40 + 3t$$

The equation gives the speed s , in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, of the car 5 seconds after it began to accelerate?

- A. 40
- B. 43
- C. 45
- D. 55

Q2

ALGEBRA

The function g is defined by $gx = 10x + 8$. What is the value of gx when $x = 8$?

- A. 0
- B. 8
- C. 10
- D. 88

Q3

GRID-IN ALGEBRA

The function f is defined by $fx = 4x$. For what value of x does $fx = 8$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

ALGEBRA

The total cost $f(x)$, in dollars, to lease a car for 36 months from a particular car dealership is given by $f(x) = 36x + 1,000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is \$ 400?

- A. \$ 13,400
- B. \$ 13,000
- C. \$ 15,400
- D. \$ 37,400

Q5

ALGEBRA

If f is the function defined by $f(x) = \frac{2x-1}{3}$, what is the value of $f(5)$?

- A. $\frac{4}{3}$
- B. $\frac{7}{3}$
- C. 3
- D. 9

Q6

ALGEBRA

In the linear function h , $h(0) = 41$ and $h(1) = 40$. Which equation defines h ?

A. $h(x) = -x + 41$

B. $h(x) = -x$

C. $h(x) = -41x$

D. $h(x) = -41$

Q7

ALGEBRA

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 39 and passes through the point $0, 0$. Which equation defines f ?

A. $f(x) = -39x$

B. $f(x) = \frac{1}{39}x$

C. $f(x) = x - 39$

D. $f(x) = 39x$

To repair a refrigerator, a technician charges \$ 60 per hour for labor plus \$ 120 for parts. Which function f represents the total amount, in dollars, the technician will charge for this job if it takes x hours?

- A. $fx = x + 120$
- B. $fx = 60x$
- C. $fx = 60x + 120$
- D. $fx = 60x - 120$